This is an atomic read-modify-write operation for the memory location comprising `*this`.

Std::Atomic_Flag_Test_and_Set Nonmember Function

Atomically sets the flag and checks whether or not it was set.

Declaration

```
bool atomic_flag_test_and_set(volatile atomic_flag* flag) noexcept;
bool atomic_flag_test_and_set(atomic_flag* flag) noexcept;
```

Effects

```
return flag->test_and_set();
```

Std::Atomic_Flag_Test_And_Set_Explicit Nonmember Function

Atomically sets the flag and checks whether or not it was set.

Declaration

```
bool atomic_flag_test_and_set_explicit(
    volatile atomic_flag* flag, memory_order order) noexcept;
bool atomic_flag_test_and_set_explicit(
    atomic_flag* flag, memory_order order) noexcept;
```

Effects

```
return flag->test_and_set(order);
```

Std::Atomic_Flag::Clear Member Function

Atomically clears the flag.

Declaration

```
void clear(memory_order order = memory_order_seq_cst) volatile
noexcept;
void clear(memory_order order = memory_order_seq_cst) noexcept;
```